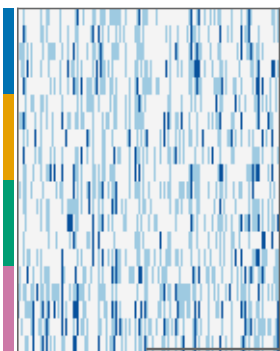


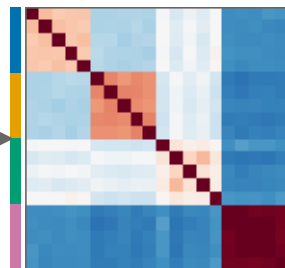
dwg: uncentred GRM, diagonal free, then whitened and centred

genotypes G



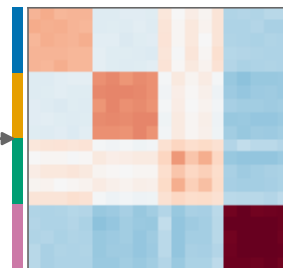
standardise,
not centred

GRM
(scaled, uncentred)



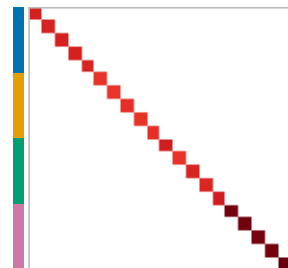
fit
 $A = L + S$

L : diag. imputed by
top 3 PCs (rank 4)



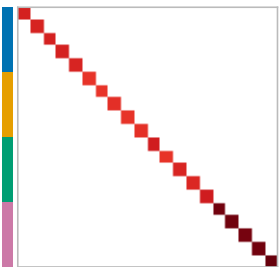
+

S : the diagonal



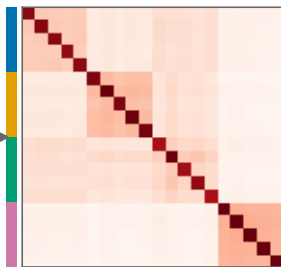
noise $v = S$

$\Sigma = \text{diag}(v)$



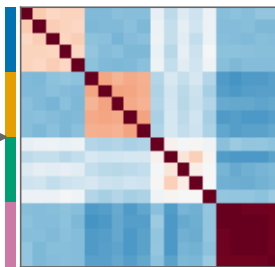
$\Sigma^{-1/2} A \Sigma^{-1/2}$

whitened: $A_{ij}/\sqrt{v_i v_j}$
(not centred: all positive)



centre

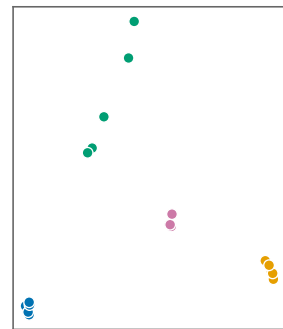
centred: $\Sigma^{-1/2} \mathbf{1}$
projected out



top k ,
 $\times \Sigma^{1/2}$

PC3

PCs: PC2 vs PC3
MXL axis $R^2 = 0.99$



PC2

● CEU ● CHB ● MXL ● YRI